Pre-registered comparison C2: ring layout vs free layout

Written 2026-10-20, before the ring-mode code exists, and committed to git before any run. Context: Reza said placement "still doesn't look great" and asked for head-symmetric arrays (Design v1.6).

Arms

- **Free:** the current solver (fast mode, target = nearest ring per v1.1, worst-pair term on), **multi-start best of 4** (WP3).
- **Ring:** ring mode, mirror-symmetric.
  - Ring counts are allocated automatically from N.
  - Per-ring elevation, spacing, rotation and **size** are optimised.
  - Bloom and sharpness are fixed at the same values as the free arm, so only the layout differs.

Both arms use the same defaults: helmet, face exclusion zones, placeholder wire, threshold 0.01.

Configurations (4)

#	Knot	β	m	N
1	T(2,3)	0.8	1	16
2	T(2,3)	0.8	1	24
3	T(2,3)	0.8	1	32
4	T(2,5)	0.8	2	24

Outcomes

All outcomes come from the common exact final report, so both arms are scored identically:

- **worst adjacent |k|**, the worst coupling among Voronoi-adjacent pairs (the v1.6 definition);
- **regularity CV**, the coefficient of variation of nearest-neighbour centre distances (lower = more even);
- **coverage**;
- **zone violations.**

Reported for information: median adjacent |k|, second-ring worst |k|, runtime.

Decision rule (applied mechanically)

Ring becomes the **default layout mode** only if **all three** hold:

1. Ring regularity CV is lower than free in **≥ 3 of 4** configurations.
2. The median over configurations of (ring worst adjacent |k|) ÷ (free worst adjacent |k|) is **≤ 1.10**, meaning ring is not more than 10% worse.
3. Ring coverage is **≥ 0.97** in all four configurations, with **zero zone violations.**

Otherwise free stays the default, and ring mode stays available as an option.

Status

Exploratory: placeholder helmet, zones and wire; these configurations only.